

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/23/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `c:\Users\mohog\hw3-mohogos`

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

Given constants * $U = 6$ km/day * $k_a = 0.55$ /day * $k_c = 0.35$ /day * $k_n = 0.25$ /day * $C_s = 10$ mg/L * $x = 50$ km

Analytic Solution Equation (ignoring photosynthesis, respiration, and benthic uptake)

$$C(x) = C_s(1 - a_1) + C_0a_1 - B_0a_2 - N_0a_3$$

Determine concentration at $x = 0$ km

$$\begin{aligned}
 a_1 &= e^{-\frac{0.55 \cdot 0}{6}} = e^0 = 1 \\
 a_2 &= \left(\frac{0.35}{0.55 - 0.35} \right) \left(e^{-\frac{0.35 \cdot 0}{6}} - e^{-\frac{0.55 \cdot 0}{6}} \right) \\
 &= 1.75 \cdot (1 - 1) = 0 \\
 a_3 &= \left(\frac{0.25}{0.55 - 0.25} \right) \left(e^{-\frac{0.25 \cdot 0}{6}} - e^{-\frac{0.55 \cdot 0}{6}} \right) \\
 &= 0.833 \cdot (1 - 1) = 0
 \end{aligned}$$

$$\begin{aligned}
 C(x) &= 10(1 - a_1) + C_0 a_1 - B_0 a_2 - N_0 a_3 \\
 C(x = 0) &= 10(1 - 1) + C_0(1) - B_0(0) - N_0(0) = C_0
 \end{aligned}$$

Determine values of C_0 , B_0 , and N_0 * Total flow: 100,000 m³/day + 10,000 m³/day = 110,000 m³/day = 110000000 L/day * Ratio: river inflow = 10/11, waste stream 1 = 1/11

$$\begin{aligned}
 C_0 &= (7.5 \text{ mg/L}) \cdot \left(\frac{10}{11} \right) + (5 \text{ mg/L}) \cdot \left(\frac{1}{11} \right) \\
 &= 7.27 \text{ mg/L} \\
 B_0 &= (5 \text{ mg/L}) \cdot \left(\frac{10}{11} \right) + (50 \text{ mg/L}) \cdot \left(\frac{1}{11} \right) \\
 &= 9.09 \text{ mg/L} \\
 N_0 &= (5 \text{ mg/L}) \cdot \left(\frac{10}{11} \right) + (35 \text{ mg/L}) \cdot \left(\frac{1}{11} \right) \\
 &= 7.72 \text{ mg/L}
 \end{aligned}$$

Total dissolved oxygen concentration, $x = 0$

$$C(x = 0) = 7.27$$

Determine a values and DO concentration at $x = 15$ km

$$\begin{aligned}
 a_1 &= e^{-\frac{0.55 \cdot 15}{6}} = 0.253 \\
 a_2 &= \left(\frac{0.35}{0.55 - 0.35} \right) \left(e^{-\frac{0.35 \cdot 15}{6}} - e^{-\frac{0.55 \cdot 15}{6}} \right) \\
 &= 1.75 \cdot (0.164) = 0.287 \\
 a_3 &= \left(\frac{0.25}{0.55 - 0.25} \right) \left(e^{-\frac{0.25 \cdot 15}{6}} - e^{-\frac{0.55 \cdot 15}{6}} \right) \\
 &= 0.833 \cdot (0.2824) = 0.235 \\
 C(x) &= 10(1 - 0.253) + C_0(0.253) - B_0(0.287) - N_0(0.235) \\
 C(x = 15) &= 10(0.747) + C_0(0.253) - B_0(0.287) - N_0(0.235)
 \end{aligned}$$

Apply previous initial concentrations

$$C(x = 15) = 10(0.747) + 7.27(0.253) - 9.09(0.287) - 7.72(0.235)$$

$$C(x = 15) = 4.89$$

Account for decay of CBOD and NBOD before mixing at 15 m

$$B_0 = 9.09 \cdot e^{-\frac{0.35 \cdot 15}{6}}$$

$$= 3.79$$

$$N_0 = 7.72 \cdot e^{-\frac{0.25 \cdot 15}{6}}$$

$$= 4.13$$

Determine values of C0, B0, and N0 after mixing with waste stream 2 * Total flow: 110,000 m³/day + 15,000 m³/day = 125,000 m³/day * Ratio: river + stream 1 = 22/25, stream 2 = 3/25

$$C_0 = (4.89 \text{ mg/L}) \cdot \left(\frac{22}{25}\right) + (5 \text{ mg/L}) \cdot \left(\frac{3}{25}\right)$$

$$= 4.90 \text{ mg/L}$$

$$B_0 = (3.79 \text{ mg/L}) \cdot \left(\frac{22}{25}\right) + (45 \text{ mg/L}) \cdot \left(\frac{3}{25}\right)$$

$$= 8.735 \text{ mg/L}$$

$$N_0 = (4.13 \text{ mg/L}) \cdot \left(\frac{22}{25}\right) + (35 \text{ mg/L}) \cdot \left(\frac{3}{25}\right)$$

$$= 7.834 \text{ mg/L}$$

Total dissolved oxygen concentration, x = 15

$$C(x = 15) = 10(0.747) + 4.90(0.253) - 8.735(0.287) - 7.834(0.235)$$

$$C(x = 15) = 4.36$$

Parameter	(x = 0 km)	(x = 15 km)
a_1	1.000	0.253
a_2	0.000	0.287
a_3	0.000	0.235
C_0	7.27	4.90
B_0	9.09	8.735
N_0	7.72	7.834

Parameter	(x = 0 km)	(x = 15 km)
C(x)	7.27	4.36

```

# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day
C_s = 10.0       # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0
N1_ws = 35.0

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0
N2_ws = 35.0

# Mixing after first waste stream
CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

# Recalibrate values at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)

```

```

a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + C0_1 * a1_15 - B0_1 * a2_15 - N0_1 * a3_15
B_15 = B0_1 * exp(-k_c * t)
N_15 = N0_1 * exp(-k_n * t)

# Mixing with second waste stream at 15 km
C0_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
B0_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
N0_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Define analytic functions
function do_analytic_1(C0, B0, N0, k_a, k_n, k_c, C_s, U)
    x1 = 0:0.5:15
    C1 = Float64[]

    for x in x1
        a1 = exp(-k_a * x / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * x / U) - exp(-k_a * x / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * x / U) - exp(-k_a * x / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
        push!(C1, C)
    end

    return x1, C1
end

function do_analytic_2(C0, B0, N0, k_a, k_n, k_c, C_s, U)
    x2 = 15:0.5:50
    C2 = Float64[]

    for x in x2
        dx = x - 15
        a1 = exp(-k_a * dx / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * dx / U) - exp(-k_a * dx / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * dx / U) - exp(-k_a * dx / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
        push!(C2, C)
    end

    return x2, C2
end

```

```

x1, C1 = do_analytic_1(CO_1, BO_1, NO_1, k_a, k_n, k_c, C_s, U)
x2, C2 = do_analytic_2(CO_2, BO_2, NO_2, k_a, k_n, k_c, C_s, U)

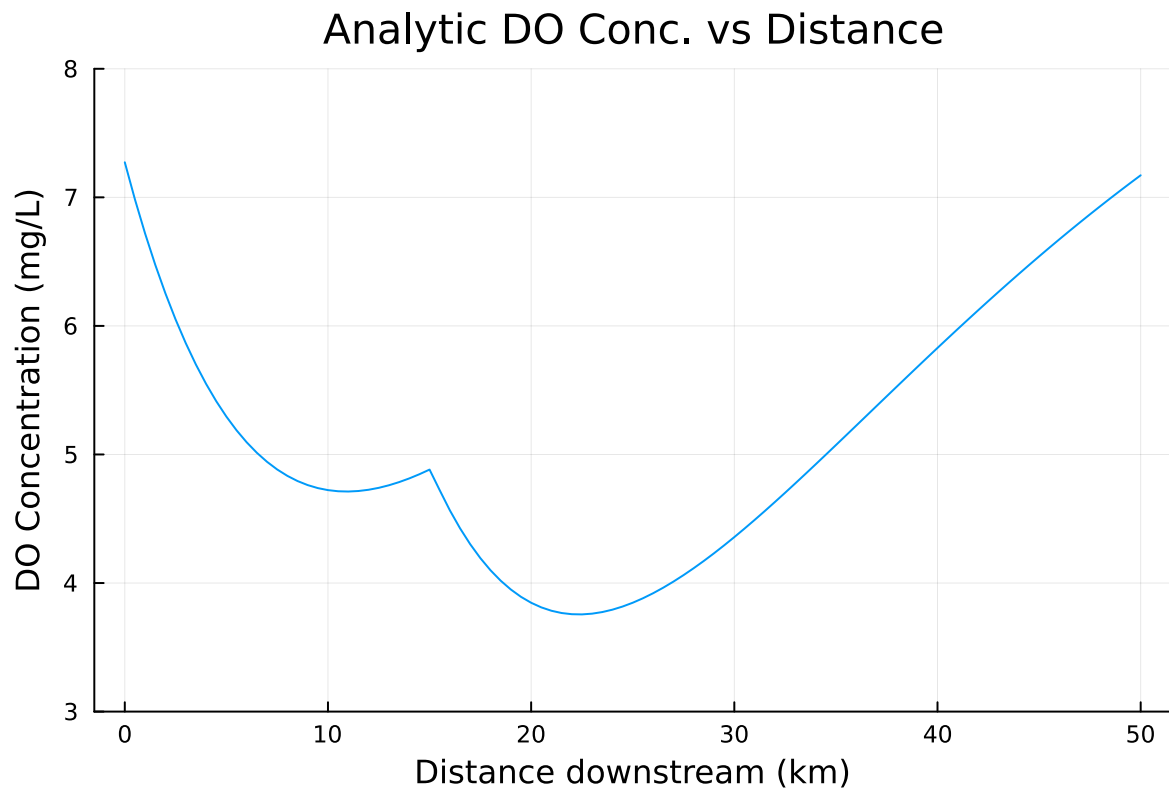
DO_distance_analytic = vcat(x1, x2[2:end])
DO_concentration_analytic = vcat(C1, C2[2:end])

min_DO = minimum(DO_concentration_analytic)
min_index = argmin(DO_concentration_analytic)
min_distance = DO_distance_analytic[min_index]

println("Minimum DO Concentration: $(round(min_DO, digits=3)) mg/L")
println("Occurs at: $(min_distance) km downstream")
plot(DO_distance_analytic, DO_concentration_analytic, xlabel = "Distance downstream (km)", y

```

Minimum DO Concentration: 3.756 mg/L
Occurs at: 22.5 km downstream



Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

```
# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day
C_s = 10.0       # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0
N1_ws = 35.0

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0
N2_ws = 35.0

# Mixing after first waste stream
CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1
```



```

# Recalibrate values at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)
a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + C0_1 * a1_15 - B0_1 * a2_15 - NO_1 * a3_15
B_15 = B0_1 * exp(-k_c * t)
N_15 = NO_1 * exp(-k_n * t)

# Mixing with second waste stream at 15 km
C0_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
B0_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
NO_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Defining distance range and DO concentration array
DO_distance_discretized = 0:0.5:50
DO_concentration_discretized = Float64[]

# Defining function
function do_numerical(L, Δx, C0, B0, NO, ka, kn, kc, Cs, U)
    n = Int(L / Δx)
    C = zeros(n + 1)
    B = zeros(n + 1)
    N = zeros(n + 1)
    C[1] = C0
    B[1] = B0
    N[1] = NO
    for i = 1:n
        B[i+1] = B[i] * exp(-kc * Δx / U)
        N[i+1] = N[i] * exp(-kn * Δx / U)
        C[i+1] = C[i] + (Δx / U) * (ka * (Cs - C[i]) - kc * B[i] - kn * N[i])
    end
    return (C, B, N)
end

L1 = 15.0
C1, B1, N1 = do_numerical(L1, Δx, C0_1, B0_1, NO_1, k_a, k_n, k_c, C_s, U)
L2 = 35.0
C2, B2, N2 = do_numerical(L2, Δx, C0_2, B0_2, NO_2, k_a, k_n, k_c, C_s, U)
DO_concentration_discretized = vcat(C1, C2[2:end])

min_DO = minimum(DO_concentration_discretized)

```

```

min_index = argmin(DO_concentration_discretized)
min_distance = DO_distance_discretized[min_index]

println("Minimum DO Concentration: $(round(min_DO, digits=3)) mg/L")
println("Occurs at: $(min_distance) km downstream")
plot(DO_distance_discretized, DO_concentration_discretized, xlabel="Distance downstream (km)")

# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day
C_s = 10.0       # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0
N1_ws = 35.0

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0
N2_ws = 35.0

# Mixing after first waste stream
CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

```

```

# Recalibrate values at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)
a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + C0_1 * a1_15 - B0_1 * a2_15 - N0_1 * a3_15
B_15 = B0_1 * exp(-k_c * t)
N_15 = N0_1 * exp(-k_n * t)

# Mixing with second waste stream at 15 km
C0_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
B0_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
N0_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Define analytic functions
function do_analytic_1(C0, B0, N0, k_a, k_n, k_c, C_s, U)
    x1 = 0:0.5:15
    C1 = Float64[]

    for x in x1
        a1 = exp(-k_a * x / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * x / U) - exp(-k_a * x / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * x / U) - exp(-k_a * x / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
        push!(C1, C)
    end

    return x1, C1
end

function do_analytic_2(C0, B0, N0, k_a, k_n, k_c, C_s, U)
    x2 = 15:0.5:50
    C2 = Float64[]

    for x in x2
        dx = x - 15
        a1 = exp(-k_a * dx / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * dx / U) - exp(-k_a * dx / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * dx / U) - exp(-k_a * dx / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
        push!(C2, C)
    end
end

```

```

    return x2, C2
end

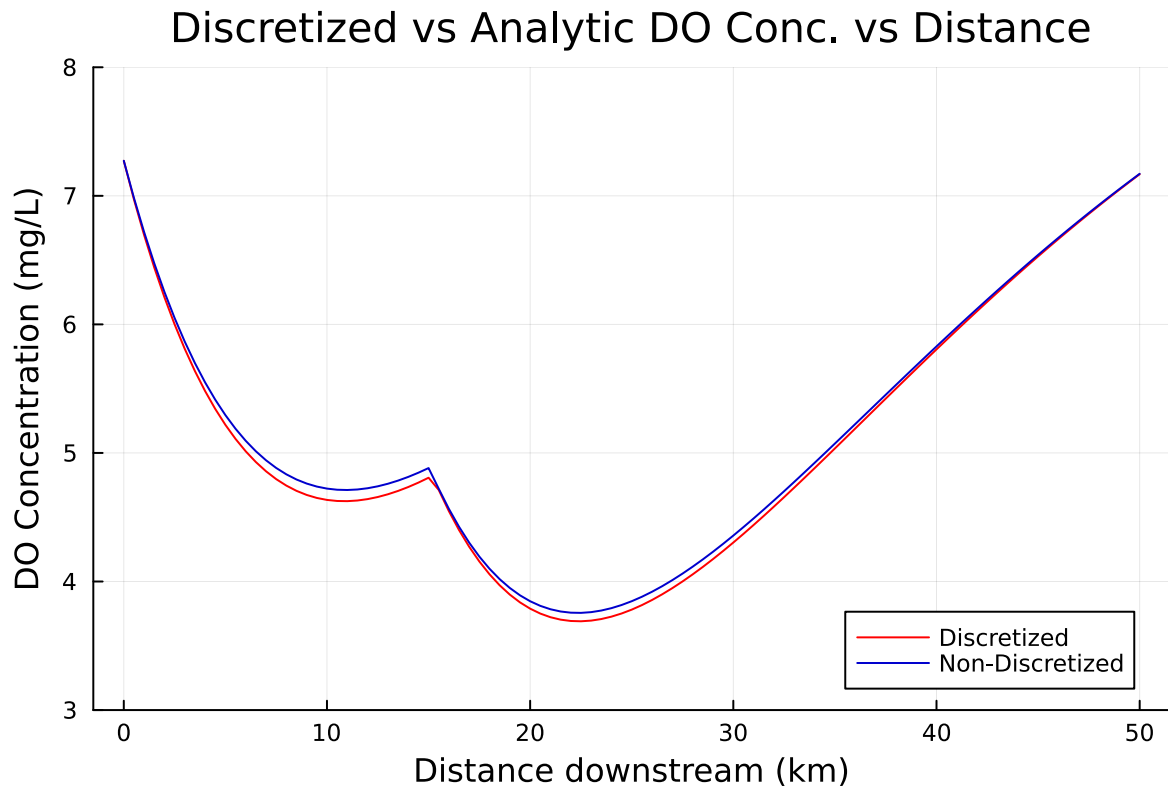
x1, C1 = do_analytic_1(CO_1, BO_1, NO_1, k_a, k_n, k_c, C_s, U)
x2, C2 = do_analytic_2(CO_2, BO_2, NO_2, k_a, k_n, k_c, C_s, U)

DO_distance_analytic = vcat(x1, x2[2:end])
DO_concentration_analytic = vcat(C1, C2[2:end])
plot!(DO_distance_analytic, DO_concentration_analytic, legend = true, linecolor=:mediumblue)

```

Minimum DO Concentration: 3.691 mg/L

Occurs at: 22.5 km downstream



Although the discretized solution identifies the minimum DO concentration as being at the same location as in the non-discretized solution - 22.5 km downstream - the minimum value has slightly decreased from 3.76 to 3.69 mg/L. The discretized solution introduces greater complexity in the calculations by making the DO concentration, CBOD, and NBOD be time-varying with every increase in the step size of 0.5 km. Because the DO concentration is being

recalculated with values that update every 0.5 km instead of a consistent numerical value for the parameter, the discretized solution predicts slightly greater oxygen consumption levels and correspondingly lower DO concentrations.

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

Applying 5% margin of safety to compliance minimum:

$$C(x) \geq 4 * 1.05 = 4.2 \text{ mg/L}$$

Approach

The following code builds on the code from 1.1, adds a variable R representing the amount of reduction such that $(1-R)*100$ is the percent removal of organic waste, multiplies this variable by both B#_ws and N#_ws (where the # represents the specific waste stream), and then solves for the minimum level of DO. I achieved the results by manually altering the value of R until the minimum DO concentration was outputted as approximately 4.200 mg/L for both scenarios. The results are shown below.

Waste 1 treated, waste 2 untreated

```
# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day
C_s = 10.0       # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
```

```

Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Add variable R representing reduction of CBOD + NBOD
R = 0.7242

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0*R
N1_ws = 35.0*R

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0
N2_ws = 35.0

# Mixing after first waste stream
CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

# Recalibrate values at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)
a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + CO_1 * a1_15 - BO_1 * a2_15 - NO_1 * a3_15
B_15 = BO_1 * exp(-k_c * t)
N_15 = NO_1 * exp(-k_n * t)

# Mixing with second waste stream at 15 km
CO_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
BO_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
NO_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Define analytic functions
function do_analytic_1(C0, BO, NO, k_a, k_n, k_c, C_s, U)
    x1 = 0:0.5:15

```

```

C1 = Float64[]

for x in x1
    a1 = exp(-k_a * x / U)
    a2 = (k_c / (k_a - k_c)) * (exp(-k_c * x / U) - exp(-k_a * x / U))
    a3 = (k_n / (k_a - k_n)) * (exp(-k_n * x / U) - exp(-k_a * x / U))
    C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
    push!(C1, C)
end

return x1, C1
end

function do_analytic_2(C0, B0, N0, k_a, k_n, k_c, C_s, U)
    x2 = 15:0.5:50
    C2 = Float64[]

    for x in x2
        dx = x - 15
        a1 = exp(-k_a * dx / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * dx / U) - exp(-k_a * dx / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * dx / U) - exp(-k_a * dx / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - N0 * a3
        push!(C2, C)
    end

    return x2, C2
end

x1, C1 = do_analytic_1(C0_1, B0_1, N0_1, k_a, k_n, k_c, C_s, U)
x2, C2 = do_analytic_2(C0_2, B0_2, N0_2, k_a, k_n, k_c, C_s, U)

DO_distance_analytic = vcat(x1, x2[2:end])
DO_concentration_analytic = vcat(C1, C2[2:end])

min_DO = minimum(DO_concentration_analytic)
min_index = argmin(DO_concentration_analytic)
min_distance = DO_distance_analytic[min_index]

println("Minimum DO Concentration: $(min_DO)) mg/L")
println("Occurs at: $(min_distance) km downstream")

```

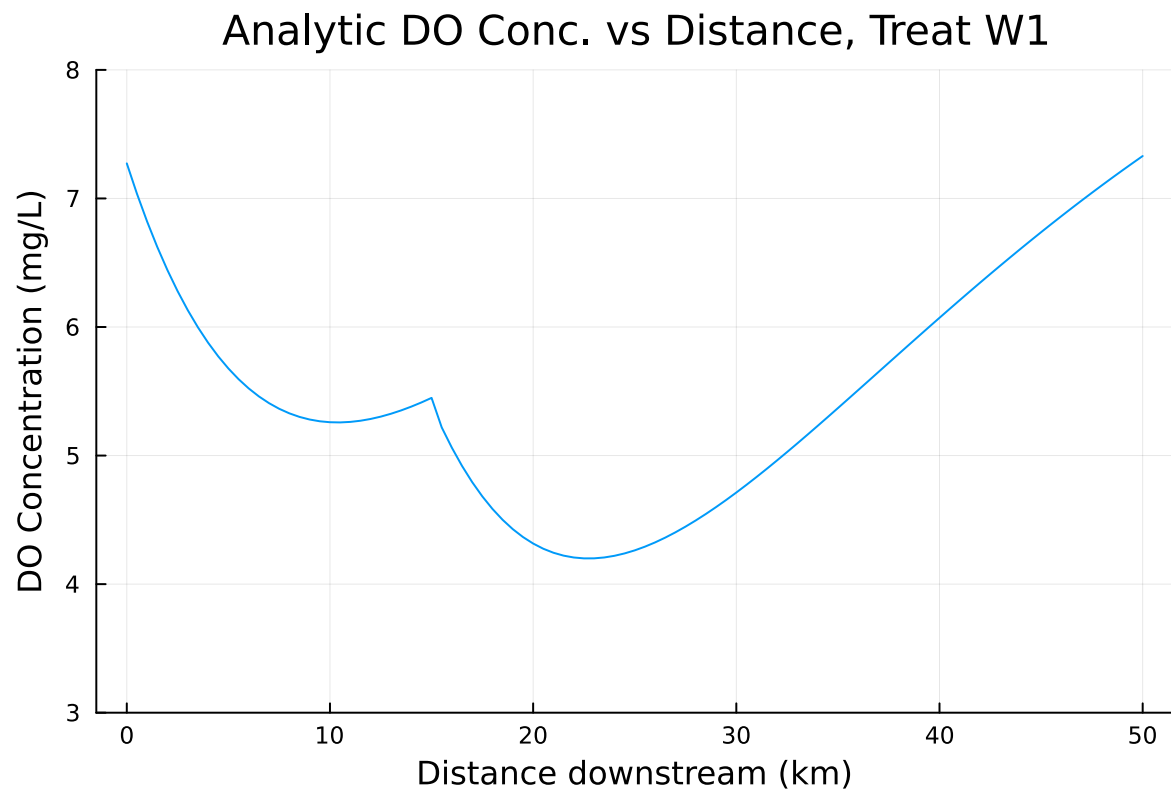
```

min_treatment = (1-R)*100
println("Minimum Level of Treatment: $(round(min_treatment, digits=2)) %")

plot(DO_distance_analytic, DO_concentration_analytic, xlabel = "Distance downstream (km)", y

```

Minimum DO Concentration: 4.200360487612862) mg/L
 Occurs at: 22.5 km downstream
 Minimum Level of Treatment: 27.58 %



Waste 2 treated, waste 1 untreated

```

# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day

```



```

C_s = 10.0          # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0
N1_ws = 35.0

# Add variable R representing reduction of CBOD + NBOD
R = 0.7825

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0*R
N2_ws = 35.0*R

# Mixing after first waste stream
CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

# Recalibrate values at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)
a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + CO_1 * a1_15 - BO_1 * a2_15 - NO_1 * a3_15
B_15 = BO_1 * exp(-k_c * t)
N_15 = NO_1 * exp(-k_n * t)

```

```

# Mixing with second waste stream at 15 km
C0_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
B0_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
NO_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Define analytic functions
function do_analytic_1(C0, B0, NO, k_a, k_n, k_c, C_s, U)
    x1 = 0:0.5:15
    C1 = Float64[]

    for x in x1
        a1 = exp(-k_a * x / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * x / U) - exp(-k_a * x / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * x / U) - exp(-k_a * x / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - NO * a3
        push!(C1, C)
    end

    return x1, C1
end

function do_analytic_2(C0, B0, NO, k_a, k_n, k_c, C_s, U)
    x2 = 15:0.5:50
    C2 = Float64[]

    for x in x2
        dx = x - 15
        a1 = exp(-k_a * dx / U)
        a2 = (k_c / (k_a - k_c)) * (exp(-k_c * dx / U) - exp(-k_a * dx / U))
        a3 = (k_n / (k_a - k_n)) * (exp(-k_n * dx / U) - exp(-k_a * dx / U))
        C = C_s * (1 - a1) + C0 * a1 - B0 * a2 - NO * a3
        push!(C2, C)
    end

    return x2, C2
end

x1, C1 = do_analytic_1(C0_1, B0_1, NO_1, k_a, k_n, k_c, C_s, U)
x2, C2 = do_analytic_2(C0_2, B0_2, NO_2, k_a, k_n, k_c, C_s, U)

DO_distance_analytic = vcat(x1, x2[2:end])
DO_concentration_analytic = vcat(C1, C2[2:end])

```

```

min_DO = minimum(DO_concentration_analytic)
min_index = argmin(DO_concentration_analytic)
min_distance = DO_distance_analytic[min_index]

println("Minimum DO Concentration: $(min_DO) mg/L")
println("Occurs at: $(min_distance) km downstream")

min_treatment = (1-R)*100
println("Minimum Level of Treatment: $(round(min_treatment, digits=2)) %")

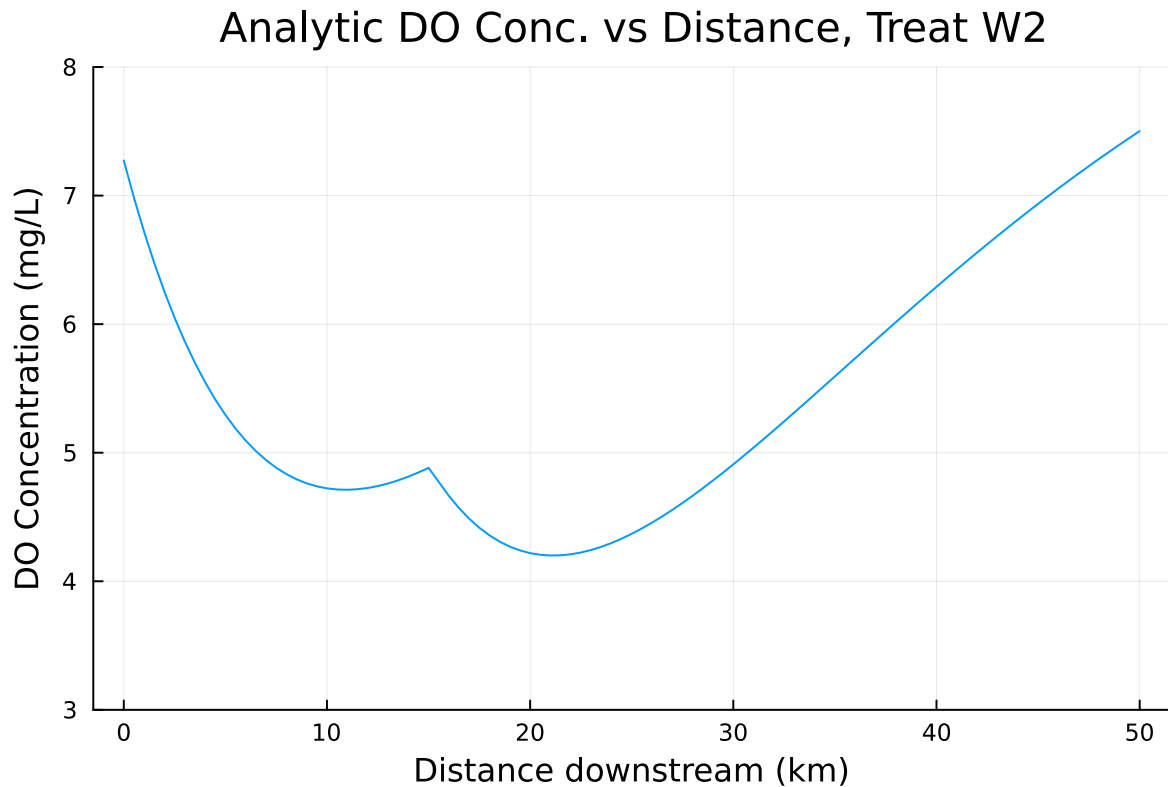
plot(DO_distance_analytic, DO_concentration_analytic, xlabel = "Distance downstream (km)", y:

```

Minimum DO Concentration: 4.2003962408062065 mg/L

Occurs at: 21.0 km downstream

Minimum Level of Treatment: 21.75 %



When only waste stream 1 is treated, a minimum treatment level of 27.58% must be applied. When only waste stream 2 is treated, a minimum level of 21.75% must be

applied.

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

From the initial graph from 1.1, we saw that the DO level did not dip below 4 mg/L (or, with the safety factor, 4.2 mg/L) until after the input from waste stream 2. Following the analysis above, waste stream 1 requires a treatment level of at least 27.58%, while waste stream 2 requires a slightly lower treatment level of 21.75%. Since the DO level does not dip below the standard with safety factor before the input from stream 2 and a lower treatment level would be expected to require fewer resources and monetary expenses in a real-world context, I would only treat waste stream 2.

That said, the two sources are 15 km apart, and the nature of the sources are not specified. Depending on the infrastructure/terrain at each location, it could end up being logistically easier to only treat waste stream 1. In the same vein, it may be more expensive to treat stream 2 compared to stream 1, which presents another consideration from an economic standpoint. Additionally, this hypothetical river could very well be located across US state boundaries, in which case minimum DO concentrations for both states should be considered as states are permitted to have stricter standards for water quality parameters than the federal level; this would directly affect the level of treatment needed. Lastly, any projected changes in waste stream inputs in the near future would also be relevant - if either stream experiences a notable increase/decrease in CBOD and NBOD in the near future due to human activity or environmental changes, this would affect the previous calculations and the resulting minimum treatment levels required to meet the DO standard.

Problem 1.5

- Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution.
- Conduct a Monte Carlo simulation of the DO concentration in the river.
- If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the regulatory standard of 4 mg/L?
- How did you decide that your Monte Carlo sample size was sufficiently large?

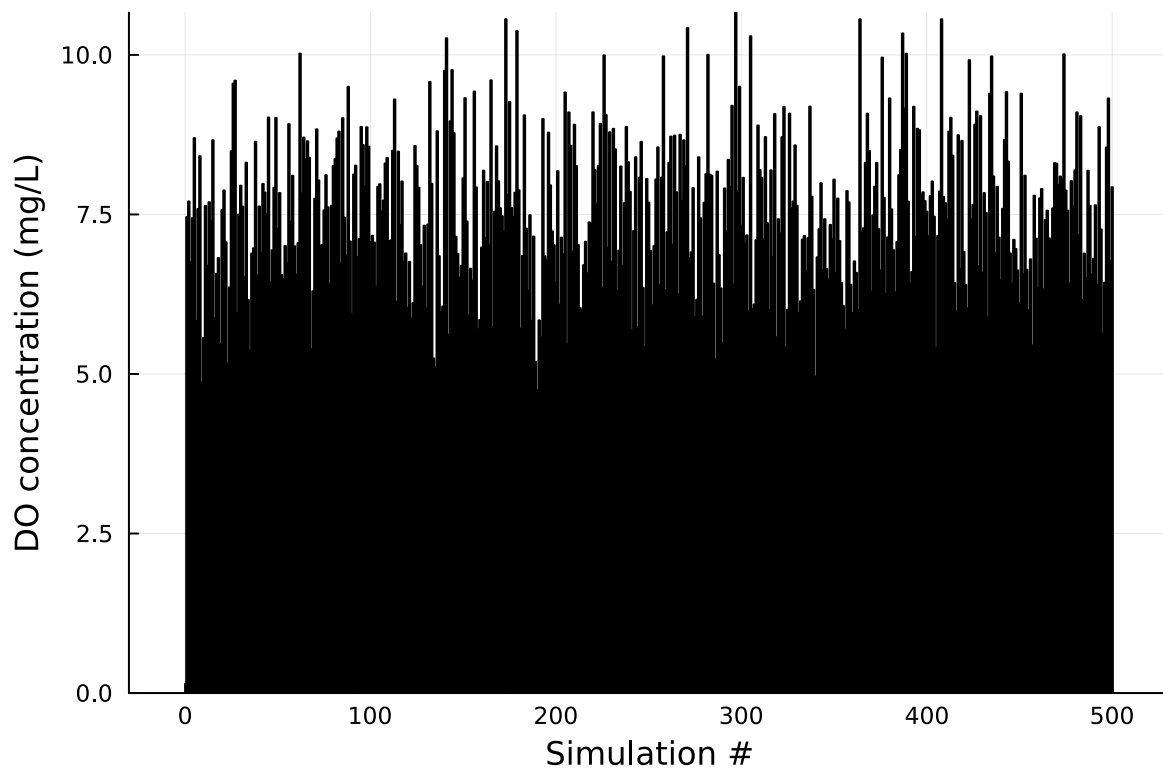
```
Random.seed!(1)
```

```
TaskLocalRNG()
```

```
# Apply LogNormal probability distribution for DO concentration of river inflow  
C_river = 7.5 #mg/L
```

```
ln_distribution = LogNormal(2.0, 0.15)  
ln_DO_river = rand(ln_distribution, 500)
```

```
bar(ln_DO_river, legend=:false, bins=500)  
ylabel!("DO concentration (mg/L)")  
xlabel!("Simulation #")
```



```
# Output Monte Carlo estimates
```

```
mean_distribution = mean(ln_DO_river)
```

```
println(" Mean Monte Carlo estimate of DO concentration at river inflow: $(mean_distribution)")
```

Mean Monte Carlo estimate of DO concentration at river inflow: 7.5898559944112485

Monte Carlo Simulation of DO Concentration in River, Both Waste Streams Untreated

```
# Constants
U = 6.0          # km/day
k_a = 0.55       # /day
k_c = 0.35       # /day
k_n = 0.25       # /day
C_s = 10.0       # mg/L

# Flows (m³/day)
Q_r = 100000.0
Q1  = 10000.0
Q2  = 15000.0

Q_m1 = Q_r + Q1
Q_total = Q_m1 + Q2

# Inflow concentrations
C_r = 7.5
B_r = 5.0
N_r = 5.0

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0
N1_ws = 35.0

# Waste stream 2
C2_ws = 5.0
B2_ws = 45.0
N2_ws = 35.0

# Account for all projections with all samples
all_profiles = Vector{Tuple{Vector{Float64}, Vector{Float64}}}()
min_DO_vals = Float64[]
min_dist_vals = Float64[]

# Take values from the LogNormal distribution above as the river input and use same equations
for C_r in ln_DO_river

    CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
    BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
```

```

NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

# Recalibrate at 15 km before mixing
t = 15 / U
a1_15 = exp(-k_a * t)
a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
C_15 = C_s * (1 - a1_15) + CO_1 * a1_15 - BO_1 * a2_15 - NO_1 * a3_15
B_15 = BO_1 * exp(-k_c * t)
N_15 = NO_1 * exp(-k_n * t)

# Mixing with second waste stream at 15 km
CO_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
BO_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
NO_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

# Calculate DO along river
x1, C1 = do_analytic_1(CO_1, BO_1, NO_1, k_a, k_n, k_c, C_s, U)
x2, C2 = do_analytic_2(CO_2, BO_2, NO_2, k_a, k_n, k_c, C_s, U)

DO_distance = vcat(x1, x2[2:end])
DO_concentration = vcat(C1, C2[2:end])

# Store each profile
push!(all_profiles, (DO_distance, DO_concentration))

# Store minimum DO
min_DO = minimum(DO_concentration)
min_index = argmin(DO_concentration)
min_distance = DO_distance[min_index]
push!(min_DO_vals, min_DO)
push!(min_dist_vals, min_distance)
end

println("Average Minimum DO Concentration: $(mean(min_DO_vals)) mg/L")
println("Average Location of Minimum DO: $(mean(min_dist_vals)) km downstream")

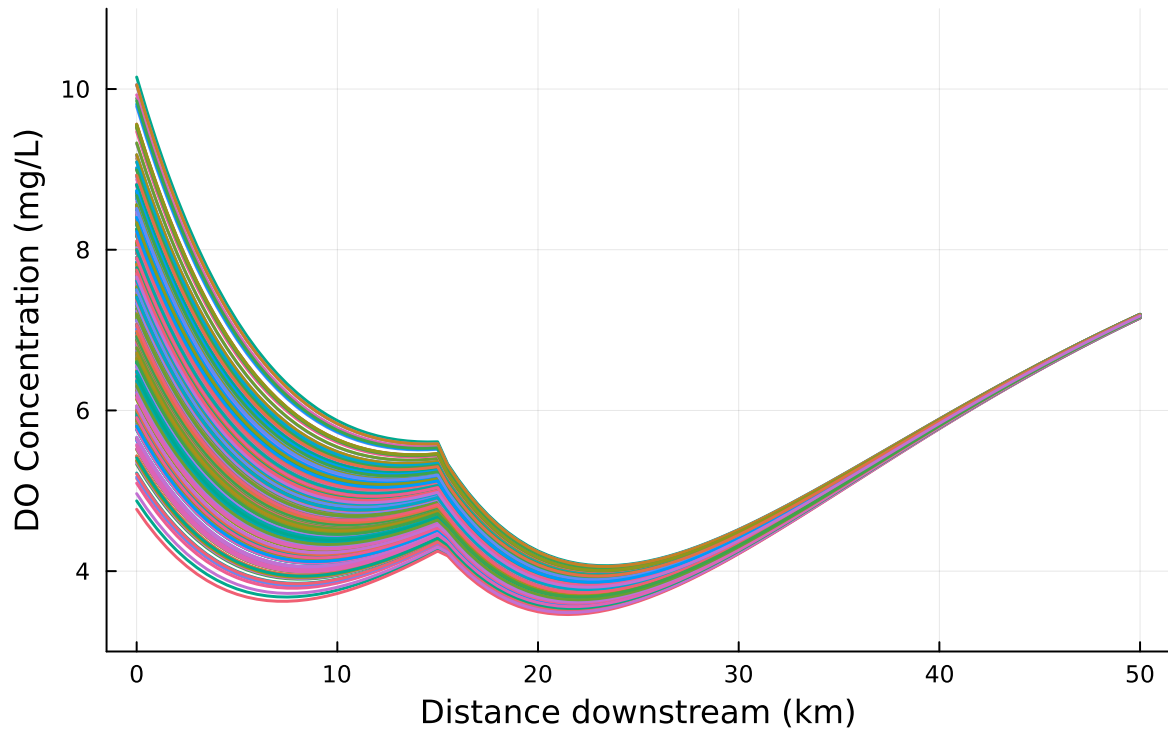
plt = plot(title="Monte Carlo Simulation, 500 Samples, Untreated", xlabel="Distance downstream", ylabel="DO Concentration (mg/L)", lw=1.5)

for (dist, conc) in all_profiles
    plot!(plt, dist, conc, lw=1.5)
end

```

```
display(plt)
```

Monte Carlo Simulation, 500 Samples, Untreated



Average Minimum DO Concentration: 3.755778975362634 mg/L

Average Location of Minimum DO: 22.347 km downstream

Monte Carlo Simulation of DO Concentration in River, Waste Stream 1 Treated

```
# Add variable R representing reduction of CBOD + NBOD
R = 0.7242

# Waste stream 1
C1_ws = 5.0
B1_ws = 50.0*R
N1_ws = 35.0*R

all_profiles = Vector[Tuple[Vector[Float64], Vector[Float64]]]()
min_DO_vals = Float64[]
```



```

min_dist_vals = Float64[]

# Take values from the LogNormal distribution above as the river input and use same equations
for C_r in ln_DO_river

    CO_1 = (Q_r*C_r + Q1*C1_ws) / Q_m1
    BO_1 = (Q_r*B_r + Q1*B1_ws) / Q_m1
    NO_1 = (Q_r*N_r + Q1*N1_ws) / Q_m1

    # Recalibrate at 15 km before mixing
    t = 15 / U
    a1_15 = exp(-k_a * t)
    a2_15 = (k_c / (k_a - k_c)) * (exp(-k_c * t) - exp(-k_a * t))
    a3_15 = (k_n / (k_a - k_n)) * (exp(-k_n * t) - exp(-k_a * t))
    C_15 = C_s * (1 - a1_15) + CO_1 * a1_15 - BO_1 * a2_15 - NO_1 * a3_15
    B_15 = BO_1 * exp(-k_c * t)
    N_15 = NO_1 * exp(-k_n * t)

    # Mixing with second waste stream at 15 km
    CO_2 = (Q_m1*C_15 + Q2*C2_ws) / Q_total
    BO_2 = (Q_m1*B_15 + Q2*B2_ws) / Q_total
    NO_2 = (Q_m1*N_15 + Q2*N2_ws) / Q_total

    # Calculate DO along river
    x1, C1 = do_analytic_1(CO_1, BO_1, NO_1, k_a, k_n, k_c, C_s, U)
    x2, C2 = do_analytic_2(CO_2, BO_2, NO_2, k_a, k_n, k_c, C_s, U)

    DO_distance = vcat(x1, x2[2:end])
    DO_concentration = vcat(C1, C2[2:end])

    # Store each profile
    push!(all_profiles, (DO_distance, DO_concentration))

    # Store minimum DO
    min_DO = minimum(DO_concentration)
    min_index = argmin(DO_concentration)
    min_distance = DO_distance[min_index]
    push!(min_DO_vals, min_DO)
    push!(min_dist_vals, min_distance)
end

println("Average Minimum DO Concentration: $(mean(min_DO_vals)) mg/L")

```

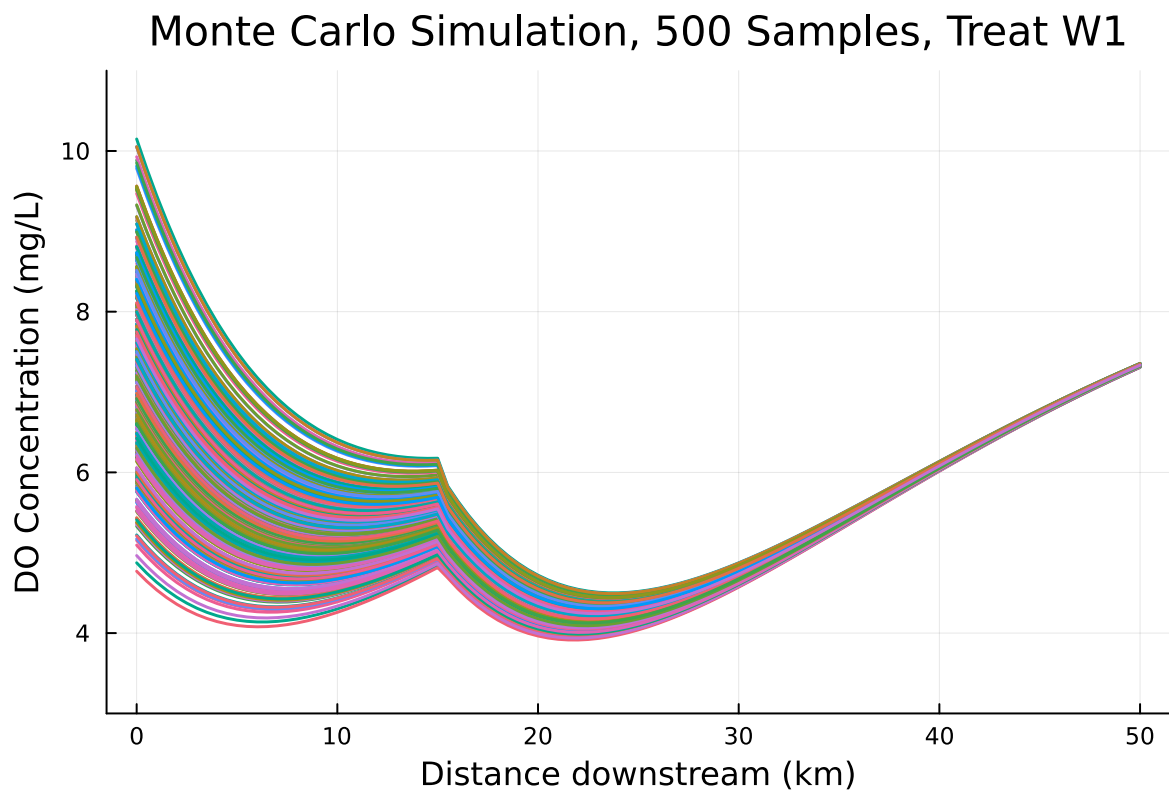
```
println("Average Location of Minimum DO: $(mean(min_dist_vals)) km downstream")

plt = plot(title="Monte Carlo Simulation, 500 Samples, Treat W1", xlabel="Distance downstream")

for (dist, conc) in all_profiles
    plot!(plt, dist, conc, lw=1.5)
end

display(plt)
```

Average Minimum DO Concentration: 4.199645548879569 mg/L
 Average Location of Minimum DO: 22.734 km downstream



Expected probability and 95% Confidence Interval

```
fails = min_DO_vals .< 4.0
p_fail = mean(fails)
```

```

n = length(fails)

z = 1.96
se = sqrt(p_fail * (1 - p_fail) / n)
lower_CI = max(0, p_fail - z * se)
upper_CI = min(1, p_fail + z * se)

println("Expected probability (DO < 4 mg/L): $(round(p_fail*100, digits=4)) %")
println("95% Confidence Interval: [$(round(lower_CI*100, digits=4)) %, $(round(upper_CI*100,

```

Expected probability (DO < 4 mg/L): 3.4 %
 95% Confidence Interval: [1.8115 %, 4.9885 %]

Applying safety factor of 5%

```

fails = min_DO_vals .< 4.2
p_fail = mean(fails)
n = length(fails)

z = 1.96
se = sqrt(p_fail * (1 - p_fail) / n)
lower_CI = max(0, p_fail - z * se)
upper_CI = min(1, p_fail + z * se)

println("Expected probability (DO < 4.2 mg/L): $(round(p_fail*100, digits=4)) %")
println("95% Confidence Interval: [$(round(lower_CI*100, digits=4)) %, $(round(upper_CI*100,

```

Expected probability (DO < 4.2 mg/L): 50.4 %
 95% Confidence Interval: [46.0174 %, 54.7826 %]

The 95% confidence interval is 1.81 - 4.99%; when the 5% safety factor is applied this interval becomes 46.02 - 54.78%. On average, there is only a 3.4% probability that the river fails to comply with the standard of 4 mg/L, however when the safety factor of 5% is applied, this probability jumps to 50.4%. This dramatic increase suggests that when only waste stream 1 is treated at the 27.58% treatment level determined in 1.3, the minimum DO is generally between 4 - 4.2 mg/L. Although this simulation does suggest that treating only waste stream 1 is effective at meeting the compliance limit, to be on the safe side and account for a 5% safety factor, it would be a better decision to either increase the treatment level applied to waste stream 1 or switch to treating waste stream 2, which requires a lower level of treatment to achieve the same standards.

From the initial bar chart showing the various generated values of the river inflow DO concentration, I chose to run 100 simulations. The values across simulations displayed a relatively consistent spread, but I increased the sample size to 500 to facilitate a greater range of initial values being accounted for. The sample size of 500 was suitable as it included a range of about 4.5 mg/L to over 10 mg/L with a sizable representation of these different values. Additionally, as I pursued a graphical approach to visualize how these varying initial values affected the DO levels for the entire 50 km stretch, using 500 samples helped show a diverse set of outcomes without overloading either the graph or my computer. I thus continued with 500 samples for the rest of the analysis.

References

List any external references consulted, including classmates.

- TA office hours
- Professor office hours
- <https://dataframes.julidata.org/stable/lib/functions/>
- <https://discourse.julialang.org/t/i-am-trying-to-initialize-a-variable-as-arr-vector-tuple-float64-vector-float/55613>